MIR MD NASIM HOSSAIN

PhD Candidate, Mechanical Engineering, NJIT

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SUMMARY

Mechanical Engineer and Ph.D. Candidate with 6+ years of experience spanning mechanical product development, CAD, prototyping, manufacturing, FEA, and engineering validation. Delivered a commercial FDM 3D printer from concept to a market-ready product and contributed to $500K+ in NSF-funded research resulting in 5+ peer-reviewed papers.

EDUCATION

Ph.D. in Mechanical Engineering Jan 2022 – Dec 2026 (Anticipated)
New Jersey Institute of Technology, Newark, NJ GPA: 4.00/4.00

B.Sc. in Aeronautical Engineering Jan 2013 – Feb 2017
Military Institute of Science and Technology, Bangladesh GPA: 3.66/4.00

TECHNICAL SKILLS

CAD & Mechanical Design: Creo (PTC), PTC Windchill, SolidWorks, Siemens NX, CATIA, AutoCAD, 3D/2D CAD, Mechanical Assemblies, GD&T, Engineering Drawings, BOMs, Design for Manufacturing (DFM)
Product Development & Validation: Mechanical Product Development, Design Verification, Prototyping, Root-Cause Analysis, Functional Testing, Design Optimization
Engineering Analysis: Finite Element Analysis (FEA), Computational Fluid Dynamics (CFD)
Simulation Software: ANSYS Fluent, COMSOL Multiphysics, In-house IBM Solver
Manufacturing & Prototyping: Sheet Metal, Welding, Metal Stamping, CNC Machining, Injection Molding, Casting, Extrusion, Additive Manufacturing (FDM), Heat Treatment
Programming: Python, MATLAB, Fortran, C/C++, G-Code, Bash/Unix, OpenMP, MPI, Git
Documentation & Tools: Technical Documentation, LaTeX, Microsoft Office

EXPERIENCE

Graduate Research Assistant Jan 2022 – Present
New Jersey Institute of Technology, Newark, NJ

- Designed 20+ complex 3D CAD models from imaging data in Creo, achieving dimensional accuracy within 5% and maintaining revision control in PTC Windchill
- Executed high-fidelity 3D analysis of complex internal flow systems using an in-house immersed boundary solver, resolving fluid interactions with over 1,500 deformable bodies
- Extended an in-house Fortran/C++ engineering solver with new physics and geometry-handling capabilities, reducing runtime by 20% through OpenMP parallelization
- Implemented 10+ post-processing modules for quantitative engineering analysis, improving data-processing efficiency by 30%
- Verified and validated solver outputs against analytical solutions, published benchmarks, and independent experimental measurements, maintaining errors below 2% across engineering simulations
- Delivered hands-on engineering laboratory training for 200+ students across 3 semesters, covering machining, CNC, GD&T, heat treatment, and mechanical testing

Design Engineer Feb 2017 – May 2021
ISON3D, Dhaka, Bangladesh

- Engineered a commercial FDM 3D printer from concept to production, delivering 2 months ahead of schedule by translating functional requirements into detailed mechanical designs
- Produced 2D/3D engineering drawings, CAD renderings, and BOMs for production, with a KeyShot product render adopted as the official product image and contributing to a 20% increase in sales
- Performed structural FEA for design verification, identifying and resolving performance discrepancies through root-cause analysis before physical testing, achieving under 5% deviation from test results
- Prototyped and validated mechanical assemblies through functional testing, resolving design issues before production while reducing development costs by 15%
- Established a product development and prototyping service, translating 100+ client concepts into CAD designs and functional 3D printed prototypes, generating $75K+/month in revenue

SELECTED PROJECTS

Desktop FDM 3D Printer Link
Commercial product, ISON3D

- Designed mechanical components of an FDM 3D printer across 3 subsystems (motion, extrusion, frame) in SolidWorks, applying DFM and component selection
- Implemented iterative mechanical design improvements, reducing assembly time by 15% while improving manufacturability and alignment accuracy

Closed-Loop Subsonic Wind Tunnel Design Link
Undergraduate Thesis, MIST

- Engineered a full closed-loop subsonic wind tunnel (750×750 mm test section) integrating 10 subsystems, designing components for sheet-metal fabrication, assembly, and manufacturability
- Optimized the contraction cone (ratio 5.76) and sized a variable-speed fan to achieve 106 m/s, producing a full BOM and assembly drawings

Fixed-Wing UAV Design Link
Undergraduate Design Competition, MIST

- Developed and assembled the complete UAV airframe in SolidWorks, including the wing, fuselage, control surfaces, and payload-dropping mechanism, resulting in an 8th place ranking among 76 teams
- Improved aerofoil design using ANSYS Fluent across 10+ angles of attack, evaluating lift, drag, and flow characteristics to increase the lift-to-drag ratio

3D Design & Analysis of Complex Flow Systems Link
PhD Research, NJIT

- Established a robust geometry-processing workflow to partition complex 3D flow domains into 150+ sections for high-fidelity, spatially resolved engineering analysis
- Quantified the impact of geometry and flow conditions on system performance, identifying up to 26% changes in apparent viscosity and 3 times variation in wall shear stress

PUBLICATIONS

All research papers listed below are supported by NSF and NIH funding.

[J3] Mir Md Nasim Hossain, Nien-Wen Hu, Liam B Walters, Ali Kazempour, Julia Withrow, Walter L. Murfee, and Peter Balogh (2025). "*Estimation of 3D Wall Shear Stress in New Blood Vessel Sprouts Using High-Fidelity Simulations*". *Biophysical Journal* (Vol. 125, Issue 5, pp. 1230–1246), Elsevier. Link

[J2] Mir Md Nasim Hossain, Nien-Wen Hu, Ali Kazempour, Walter L. Murfee, and Peter Balogh (2024). "*Hemodynamic Characteristics of a Tortuous Microvessel Using High-Fidelity Red Blood Cell Resolved Simulations*". *Microcirculation* (Vol. 31, Issue 7), Wiley. **[Top Cited Article 2024]** Link

[J1] Mir Md Nasim Hossain, Nien-Wen Hu, Maram Abdelhamid, Simerpreet Singh, Walter L Murfee, Peter Balogh (2023). "*Angiogenic Microvascular Wall Shear Stress Patterns Revealed Through Three-dimensional Red Blood Cell Resolved Modeling*". *Function* (Vol. 4, Issue 6), Oxford University Press (OUP). Link

Full publication list: Google Scholar (5 journal articles, 1 book chapter, 10+ conference presentations)

SELECTED PRESENTATIONS

Presented at 10+ national conferences (oral and poster); two are highlighted below:

[I2] DFD Interact Session, APS Division of Fluid Dynamics Annual Meeting, Houston, TX (Nov. 2025). Selected among the top 20% of abstracts by the session chair.

[I1] Invited Oral Presentation, Emerging Topics in Microcirculation (Opening Session), Microcirculatory Society Pre-Meeting, APS Physiology Summit (Apr. 2025).

AWARDS & RECOGNITION

Award	HPCwire Readers' Choice Award 2025 — Best Use of HPC in Life Sciences
Media Coverage	Featured in NSF ACCESS News and Medical Xpress; *Function* journal cover